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Your review report

Manuscript

Deep Learning for Lesion Classification in CT Imaging: A Systematic Literature Review

Feedback for the author(s)

Review file(s)

Deep LearningCT_I.docx

Comments to the author(s)

Manuscript Title: Deep Learning for Lesion Classification in CT Imaging A Systematic Literature Review

Recommendation: Major Revision
1. Summary of the Study
The authors review 57 studies from 2020 to 2025 on deep learning based lesion classification in CT across multiple organs, following PRISMA. They argue that CNN models remain dominant, hybrid and Transformer architectures generalize better, and data scarcity plus limited external validation remain barriers.
The strengths of this study include:
i. The multi-organ scope is a genuine contribution over single-organ reviews.
ii. The architectural taxonomy (Fig. 2) and the critique of evaluation metrics (Sec. 3.4.3) are well developed.
iii. The use of a modified Downs and Black checklist for quality assessment is appropriate (Sec. 2.5).
2. Major Comments
A. Low-Dose CT (LDCT) depth analysis is highlighted in the abstract and introduction due to radiation reduction benefits, the analysis does not deeply address how models manage the inherent elevated noise and signal degradation unique to real-world LDCT compared to standard CT datasets.
Recommendation expands section 4 (Discussion) to explicitly analyze the trade-offs between image noise, spatial resolution, and deep feature

representation under low-dose conditions.
B. Metric interpretation and class Imbalance the paper notes that metrics like AUC and accuracy are often overemphasized. However, in heavily imbalanced clinical datasets, overall accuracy can obscure poor sensitivity toward rare malignant classes.
Recommendation elaborates in section 4.3 (RQ4) on why clinically oriented metrics such as sensitivity at specified false positive rates (FROC) and F1-scores per class should be prioritized in future benchmarks.
C. Discussion on model interpretability and clinical trust section 4 highlights feature attribution methods Grad-CAM, SHAP, and LIME). However, these methods only show spatial localization, not true morphological reasoning.

Recommendation briefly discusses the transition from simple heatmaps toward mechanistic, interpretable AI and radiomic integrated decision-making models.
D. Inductive bias and data scarcity in transformer models that pure Transformer models lack the strong spatial inductive biases inherent to traditional convolutional layers, making them data-hungry and prone to severe overfitting when trained on small-scale, single-center medical imaging cohorts unless heavily regularized or pretrained.
Recommendation expands the architectural critique to address how effectively the reviewed studies bridge this gap, e.g., through hybrid local-global designs such convolutional patch embeddings combined with self-attention.
E. Computational efficiency and scalability further deep learning pipeline optimization the computational overhead associated with multi-stage training (such as pretraining, self-supervised distillation, and adversarial domain adaptation) introduces major reproducibility barriers across different clinical institutions.
Recommendation provides a critical evaluation regarding the practical feasibility of deploying such heavy (end-to-end) pipelines in standard clinical environments with limited compute resources.

3. Recommendation
The manuscript addresses a relevant gap and has clear scientific value, but the five analytical issues above A–E should be addressed before it can be considered for publication.

Confidential feedback for the Editor

Your recommendation	• ***Return for major revision***
Is the study design appropriate to answer the research question (including the use of appropriate controls), and are the conclusions supported by the evidence presented?	• ***No, but these points can be addressed with revisions***
Comments	***The design is partially appropriate (PRISMA-based SLR, aligned RQs), but several conclusions are not fully supported by the evidence: no structured quantitative synthesis, high heterogeneity without stratification, limited external-***

	validation evidence, and underdeveloped LDCT analysis. Conclusions on architectural superiority, generalizability, and LDCT performance need more cautious wording and stronger synthesis.
Are the methods sufficiently described to allow the study to be repeated?	• ***Yes***
Is the use of statistics and treatment of uncertainties appropriate?	• ***No***
Comments	***Statistical treatment is insufficient: no structured quantitative synthesis, no confidence intervals or heterogeneity measures, and the averages in Figs. 3–4 lack uncertainty estimates. No risk-of-bias or sensitivity analysis is reported. The authors should either add a structured synthesis with uncertainty reporting or justify the narrative approach and discuss heterogeneity and publication bias.***
Has guidance been provided on how overstated claims should be rewritten?	• ***No , I have not***
Comments	***The review does not provide explicit guidance on how overstated claims should be rewritten. Several claims remain insufficiently supported, e.g., Transformer models showing superior contextual learning or hybrid (Transformer) models generalizing better, yet no concrete suggestions are offered to temper or qualify them. The authors should identify overstated claims and propose specific evidence-aligned rewording for each.***
Is the presentation of the work clear?	• ***Yes***
Are the images in this manuscript (including electrophoretic gels and blots) free from apparent manipulation?	• ***Yes***

Confidential comments to the Editor

This is a very good manuscript addressing a relevant gap with a valuable multi-organ perspective. No ethical concerns, conflicts of interest, or image manipulation issues were identified.

It still needs some additions to become a complete structured quantitative synthesis, a risk-of-bias assessment, and more cautious wording of claims on architectural superiority, generalizability, and LDCT, plus per-database search details and protocol registration. These are constructive improvements, addressable without new experiments. I recommend major revision and re-review by the same reviewers in Round 2. With these additions, the manuscript would be a strong and complete contribution.